



Mata Kuliah : Pemrograman Berorientasi Object (PBO)  
Program Studi : D4 – Sistem Informasi Bisnis / D4 – Teknik Informatika  
Semester : 3

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Jobsheet Ke- : W02  
Github : <https://github.com/bwx1y/PBO-Nedy-Nugroho.git>

## Laporan Jobsheet

### Soal Praktikum

| Langkah | Jawaban/Deskripsi                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <div><div>Jawaban - 1</div><pre>public class MainDragon {<br/>    public static void main(String[] args) {<br/>        Dragon dragon1 = new Dragon(0, 0, 1);<br/>        Dragon dragon2 = new Dragon(0, 0, 1);<br/><br/>        System.out.println("=== DRAGON 1 ===");<br/>        dragon1.move(5);<br/>        dragon1.printStatus();<br/><br/>        dragon1.changeDirection(4);<br/>        dragon1.move(3);<br/>        dragon1.printStatus();<br/><br/>        System.out.println("\n=== DRAGON 2 ===");<br/>        dragon2.printStatus();<br/><br/>        dragon2.changeDirection(4);<br/>        dragon2.move(2);<br/><br/>        dragon2.changeDirection(3);<br/>        dragon2.move(4);<br/>        dragon2.printStatus();<br/>    }<br/>}</pre></div> |



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```
Jawaban - 1

import java.time.temporal.ValueRange;

public class Dragon {
    private int x;
    private int y;
    private int direction;

    public Dragon(int x, int y, int direction) {
        this.x = x;
        this.y = y;
        this.direction = direction;
    }

    public void changeDirection(int value) {
        ValueRange range = ValueRange.of(1, 4);
        boolean valid = range.isValidValue(value);

        if (value ≥ 1 && value ≤ 4) {
            this.direction = value;
        } else {
            System.out.println("Arah tidak valid! Harus 1-4.");
        }
    }

    public void move(int steps) {
        switch (this.direction) {
            case 1:
                this.y += steps;
                break;
            case 2:
                this.x += steps;
                break;
            case 3:
                this.y -= steps;
                break;
            case 4:
                this.x -= steps;
                break;
            default:
                System.out.println("Tidak di temukan");
        }
    }

    public void printStatus() {
        System.out.printf("Posisi (x,y): (%d, %d) %n", this.x, this.y);

        System.out.println("Arah: " + switch (this.direction) {
            case 1 → "Atas";
            case 2 → "Kanan";
            case 3 → "Bawah";
            case 4 → "Kiri";
            default → "Tidak di temukan";
        });

        System.out.println("=".repeat(100));
    }
}
```



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Apa yang terjadi jika method `move()` dipanggil persis setelah objek diinstansiasi? Ke mana objek berpindah? Perbaiki kode program untuk mengatasi masalah tersebut.

cara mengatasi nya adalah dengan cara memberikan default value 1 atau kalau tidak bisa memberikan contruktor yang akan memberikan devault value contoh dengan ini

```
code

public Dragon() {
    this.x = 0;
    this.y = 0;
    this.direction = 1;
}

public Dragon(int x, int y, int direction) {
    this.x = x;
    this.y = y;

    if (direction ≥ 1 && direction ≤ 4) {
        this.direction = direction;
    } else {
        System.out.println("Arah awal (" + direction + ") tidak valid! Default diatur ke 1 (Atas).");
        this.direction = 1;
    }
}
```